

DELTA MARSH, MB, Canada

WMO: 715630

Lat: 50.183N

Lon: 98.382W

Elev: 248

StdP: 98.38

Time Zone: -6.00 (NAC)

Period: 97-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.7	(c) -28.2	(d) -33.9	(e) 0.2	(f) -30.4	(g) -31.4	(h) 0.2	(i) -28.1	(j) 11.6	(k) -14.4	(l) 10.3	(m) -12.1	(n) 3.1	(o) 250	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.3	21.8	27.5	20.5	26.0	19.6	23.6	27.7	22.1	25.9	20.8	24.4	4.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.4	26.4	20.7	15.8	24.7	19.4	14.6	23.3	72.0	28.3	66.0	26.0	61.3	24.4	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.6	8.5	-34.8	32.6	2.7	1.5	-36.8	33.7	-38.4	34.6	-39.9	35.5	-41.9	36.6
			-34.9	25.8	2.7	2.0	-36.9	27.2	-38.5	28.4	-40.0	29.6	-41.9	31.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-15.4	-13.1	-6.4	3.9	10.8	16.2	19.5	18.0	13.0	5.4	-2.9	-12.0
	DBStd	13.42	7.82	7.72	7.26	5.57	4.57	3.68	2.91	3.32	4.09	4.34	5.98	7.28
	HDD10.0	3423	786	648	507	196	46	2	0	0	16	154	387	682
	HDD18.3	5657	1044	881	766	434	236	83	22	47	166	402	637	940
	CDD10.0	930	0	0	0	12	70	188	294	248	107	11	0	0
	CDD18.3	122	0	0	0	0	2	20	58	37	6	0	0	0
	CDH23.3	1099	0	0	0	5	58	179	466	311	72	8	0	0
	CDH26.7	244	0	0	0	0	13	44	103	72	11	1	0	0
Wind	WSAvg	4.0	3.9	3.9	3.8	4.1	4.4	3.9	3.2	3.7	4.1	4.4	4.4	4.0
Precipitation	PrecAvg	515	21	17	29	34	66	89	68	56	45	39	27	28
	PrecMax	702	48	46	67	79	153	146	119	129	100	85	118	49
	PrecMin	416	7	0	5	9	21	32	18	16	16	6	2	8
	PrecStd	70	10	13	18	18	36	34	29	31	22	22	26	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.4	13.3	23.9	28.3	30.8	31.5	31.3	28.0	23.7	14.3	4.7
		MCWB	2.0	2.2	7.9	13.7	15.2	22.1	24.6	23.9	19.3	14.2	8.3	2.1
	2%	DB	1.1	2.3	7.9	19.7	24.6	27.8	29.3	28.6	25.4	18.5	10.2	2.2
		MCWB	-0.1	1.0	4.8	10.9	13.9	19.6	23.0	21.7	18.4	11.9	5.6	0.7
	5%	DB	-1.3	0.6	5.0	16.4	21.7	25.2	27.5	26.7	22.9	15.1	6.9	-0.4
		MCWB	-2.2	-0.1	2.8	9.6	13.5	18.0	21.5	20.7	17.2	10.4	4.2	-1.3
	10%	DB	-3.8	-1.7	2.7	12.9	18.7	23.1	26.0	24.7	20.2	12.3	4.6	-2.1
		MCWB	-4.5	-2.5	1.3	8.1	12.7	17.4	20.5	19.9	15.8	8.8	2.7	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.5	8.2	14.2	17.7	23.9	25.7	25.5	21.1	14.9	8.7	2.5
		MCDB	3.7	3.8	11.7	20.9	23.6	28.2	30.1	29.2	25.8	22.9	13.9	4.2
	2%	WB	-0.1	1.1	4.7	11.8	15.7	21.3	24.1	23.2	19.3	12.7	6.1	0.8
		MCDB	0.8	1.9	7.7	18.8	20.8	25.2	28.1	27.2	23.4	17.2	9.0	2.2
	5%	WB	-2.1	0.0	2.9	9.9	14.4	19.6	22.6	21.6	17.9	10.7	4.2	-1.3
		MCDB	-1.3	0.8	4.8	15.7	19.3	22.9	26.4	25.3	21.7	14.5	6.8	-0.4
	10%	WB	-4.5	-2.4	1.4	8.3	13.2	18.1	21.3	20.3	16.5	9.1	2.8	-2.7
		MCDB	-3.8	-1.6	2.6	12.9	17.9	22.3	24.8	23.9	19.7	11.9	4.6	-2.1
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.1	10.3	10.6	10.3	9.7	10.6	10.9	11.2	9.8	8.6	8.4
		MCDBR	10.0	9.3	12.1	16.9	16.8	13.8	13.2	14.0	16.1	16.4	12.9	10.2
		MCWBR	8.9	8.0	8.6	9.5	8.3	7.1	7.6	8.4	9.6	9.9	8.7	8.9
	5% WB	MCDBR	10.0	9.0	11.4	15.6	12.9	11.3	12.2	11.9	13.3	14.8	12.0	9.9
		MCWBR	8.9	7.7	8.4	9.1	7.4	6.8	7.6	7.8	8.6	9.3	8.3	8.6
Clear-Sky Solar Irradiance	taub	0.255	0.276	0.300	0.354	0.374	0.378	0.390	0.382	0.347	0.299	0.266	0.246	
	taud	2.284	2.239	2.330	2.343	2.350	2.372	2.357	2.372	2.455	2.557	2.452	2.327	
	Ebn at Noon	795	864	909	889	883	880	862	851	849	844	788	756	
	Edh at Noon	67	92	101	113	118	116	116	109	90	68	57	55	
All-Sky Solar Radiation	RadAvg	1.38	2.57	3.92	5.07	5.54	5.88	6.23	5.15	3.72	2.21	1.21	0.98	
	RadStd	0.11	0.17	0.30	0.35	0.48	0.41	0.27	0.38	0.29	0.29	0.17	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	-0.56	-0.78	N/A	N/A	N/A	N/A

Nomenclature: See separate page